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| **20ME5O01 - WASTE TO ENERGY CONVERSION** | | | | |

**COURSE OBJECTIVES**
1. To enable students to understand of the concept of waste to energy.
2. To link technical and management principles for production of energy from waste.
3. To learn about the best available technologies for waste to energy.
4. To facilitate the students in developing skills in the decision making process.

**COURSE OUTCOMES:** Students are able to
**CO1:** Describe of the concept of waste to energy, classifications and principles. [K2]
**CO2:** Explain management principles for production of energy from waste. [K2]
**CO3.** Explain the best available technologies for waste to energy. [K2]
**CO4:** Describe the waste to energy options landfill gas, and energy from plastics.[K2]
**CO5:** Apply the knowledge in planning and operations of waste to energy plants [K3]

**UNIT-I : INTRODUCTION**
Waste - types of waste, Principles of waste management, Waste utilization, Waste management hierarchy, 3R Principle of Reduce, Reuse and Recycle, Waste as a resource, alternate energy source.

**UNIT-II: WASTE SOURCES & CHARACTERIZATION**
Source of waste, Waste production in different sectors such as domestic, industrial, agriculture, postconsumer waste etc, Waste management tools and techniques for reducing waste segregation and scientific disposal, Characterization of waste for energy utilization, Waste selection criteria.

**UNIT-III: TECHNOLOGIES FOR WASTE TO ENERGY**
Energy biochemical conversion – energy production from organic waste through anaerobic digestion, fermentation, Thermo-chemical conversion – combustion, incineration, heat recovery, pyrolysis, gasification, plasma arc technology, other newer technologies, Case studies.

**UNIT-IV: WASTE TO ENERGY OPTIONS**
Waste to energy options - landfill gas, methane emission, collection and recovery, Refuse Derived Fuel (RDF), Fluff, Briquettes, Pellets, Alternate Fuel Resource (AFR) – production and use in cement plants, Energy from plastic wastes, Non-recyclable plastic wastes for energy recovery, Energy recovery from wastes and optimization of its use, Energy analysis.

**UNIT-V: WASTE TO ENERGY PLANTS & ENVIRONMENTAL IMPLICATIONS**
Wasteto Energy Plants: Waste management activities – collection, segregation, transportation and storage requirements, Location and Site of waste to energy plants.
EnvironmentalImplications: Environmental impact of waste to energy, Safety and environmental standards, Savings on non-renewable fuel resources, Carbon credits and its types.

**Text Books:**
1. Marc Rogoff Francois Screve, Waste-to-Energy, 3rd Edition, William Andrew, 2019.
2. B.T. Nijaguna, Biogas Technology, 1st Edition, New Age International Pvt. Ltd, 2002.

**Reference Books:**
1. Vishal Prasad, BarkhaVaish, Advances in Waste-to-Energy Technologies, 1st Edition, CRC Press, 2019.
2. Dev Vrat Kamboj, Manoj Kumar Solanki, Waste to Energy: Prospects and Applications, 1st Edition, Springer, 2021.
3. P. Jayarama Reddy, Energy Recovery from Municipal Solid Waste by Thermal Conversion Technologies, 1st Edition, CRC Press/ Balkema, 2016.

**Web Reference:**
1. https://archive.nptel.ac.in/courses/103/107/103107125/